

Escalating SSRF in a Vulnerable Jira Instance to RCE via Docker Engine API

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@username, Andmp | A blog about infosec, bug hunting and more!.

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Content

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Originally posted by me on Reddit. This is just a repost of what I wrote there on /r/netsec, and formatting is a bit awkward. Due to some glitch I found out it couldn't be published there.

Awhile back, I posted on [Twitter](#) about achieving Remote Code Execution in Jira instances deployed over Docker.

It's a common sight these days, to come across internal bug trackers of large companies who simply don't firewall their internal network. As a result, one can achieve varying impacts, but *I believe, sky is the limit ;)*

Orangetsai has previously demonstrated some *exceptional cases of acheiveing RCE via SSRF based vulnerabilities*, which further motivated me to research into this topic.

Upon delving deeper, I found out that, *a huge number of Jira instances were exposed publically which itself is thought provoking and tempted me to look further into ways in which I could exploit it.*

- I was able to earn a couple bug bounties along the way on *BugCrowd* by successfully demonstrating XSS, with cookies scoped to parent domains in those cases allowing me to execute a full fledged ATO attack via Reflected XSS using SSRF in the Vulnerable Jira instance.

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The next considerable impact was *some critical information disclosure, which happened only at times, when you passed on a wrong or, malformed URL in your request* and Jira

gave you an *interesting stack trace containing more internal information*.

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Only in two counts of cases could I *leverage this to penetrate into the intranet of the concerned company* (which was interesting alone) and find some internal vulnerable software. The impact kept on increasing at each step as I progressed further and this was no more a simple report which most security researchers are tempted to make viz. Reflected XSS, SSRF and Exposed Jira Panels. Hence, I got something more to put in my security report.

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As I kept digging deeper, I came across a *Jira instance which was deployed over Docker*. Ask how? I put together a simple Python script that would keep hitting the *consumerUri* parameter with different payloads like `:::1`, `localhost`, and the likes, also with some deliberately filthy payloads just to exfiltrate some information. The endpoint is `https://[Jira-server host.tld]/jira/plugins/servlet/oauth/users/icon-uri?consumerUri=`

So, while monkey testing, and fuzzing I finally constructed this payload - `https://[Jira-server-host.tld]/jira/plugins/servlet/oauth/users/icon-uri?consumerUri=http://[:::1]:2375/containers/json`

Bam! We got some sensitive *docker credentials* stored as environment variables through an unauthenticated request to the Docker Engine API via an *SSRF vulnerability in that Jira instance* and are now in a position to conclude *we performed an RCE in an internal network where practically no XSS would hold that great an impact!* This could be rare but a notable case.

Do you have some ideas on how to exploit this further? Do post your ideas on this!